

Evolution

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1 Descent with Modification

1.1 Darwinian Evolution

Core Concept 1.1: Evolution

Evolution is defined as **descent with modification**, a process through which populations of organisms change over successive generations, accumulating heritable variations that distinguish descendant lineages from their ancestral forms.

Core Concept 1.2: Darwin & Natural Selection

Charles Darwin's seminal observations during the voyage of the HMS *Beagle*, particularly his documentation of morphological variation among finch populations across the Galápagos Islands, provided crucial empirical foundations for the theory of evolution by **natural selection**.



Figure 1: Galápagos island finch populations with differently evolved beak shapes

Central to this theory is the concept of **adaptations**—inherited characteristics that enhance organismal survival and reproductive success within specific environmental contexts. **Natural selection** operates as the mechanism through which individuals possessing such adaptive traits exhibit higher rates of survival and reproduction compared to peers lacking those traits.

*This process differs fundamentally from **artificial selection**, wherein humans deliberately breed individuals possessing desired characteristics, as exemplified by domestication programs in agriculture and animal husbandry (e.g., crop plants, domestic dogs).*

Darwin's theory of natural selection is predicated upon the following observations and inferences.

- (a) **Observation 1:** Members of a population exhibit variation in their inherited traits.
- (b) **Observation 2:** All species possess the capacity to produce more offspring than the environment can sustain, resulting in a mortality rate whereby many offspring fail to survive and reproduce.
- (c) **Inference 1:** Individuals whose inherited traits confer a higher probability of survival and reproduction in a given environment tend to produce more offspring than other individuals.
- (d) **Inference 2:** This differential reproductive success among individuals will result in the accumulation of favorable traits within the population over successive generations.

As such, natural selection operates as a process in which individuals bearing certain heritable traits survive and reproduce at higher rates than other individuals as a direct consequence of those traits. Over successive generations, natural selection increases the frequency of **adaptations** that confer advantages within a given environment. When environmental conditions change, or when individuals colonize novel habitats, natural selection may drive adaptation to these new conditions, potentially culminating in **speciation** (Core Concept 3.1).

It's critical to understand that evolution occurs within populations (not individuals).

1.2 Evidence of Evolution

Core Concept 1.3: Evidence of Evolution

Evolutionary theory is supported by extensive and convergent lines of scientific evidence.

- (a) **Direct Observations of Evolutionary Change:** Contemporary evolutionary processes are directly observable in natural and experimental populations, as exemplified by the evolution of drug-resistant bacterial strains (such as MRSA strains).
- (b) **Homology:** Structural, developmental, and molecular similarities resulting from common ancestry provide compelling evidence for evolutionary relationships.
- **Homologous structures** are anatomical features that represent variations upon a structural theme inherited from a *common ancestor*, such as the underlying skeletal architecture of mammalian forelimbs, avian wings, cetacean flippers, and human arms.

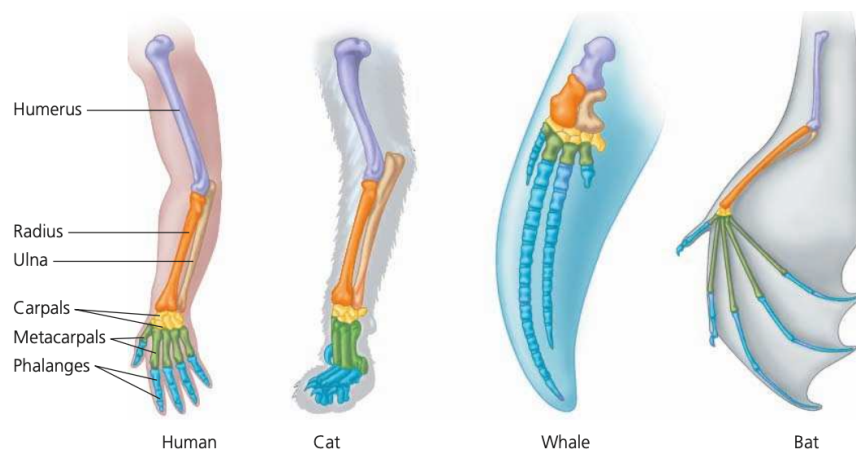


Figure 2: Homologous structure

- **Vestigial structures** are remnants of anatomical features that served functional roles in *ancestral organisms* but have been reduced or modified in descendant lineages, as exemplified by the vestigial pelvic and femoral bones retained in snake species.

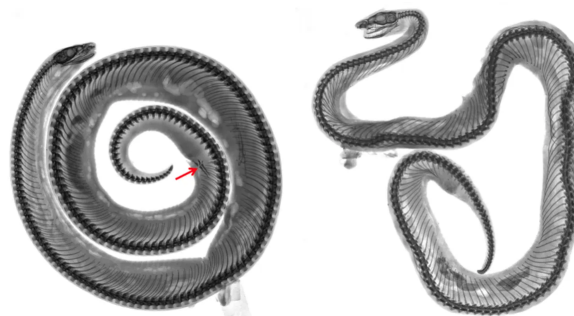


Figure 3: Vestigial structure

- **Evolutionary trees (phylogenies)** are diagrammatic representations of evolutionary relationships among groups of organisms, constructed on the basis of homologous character states.

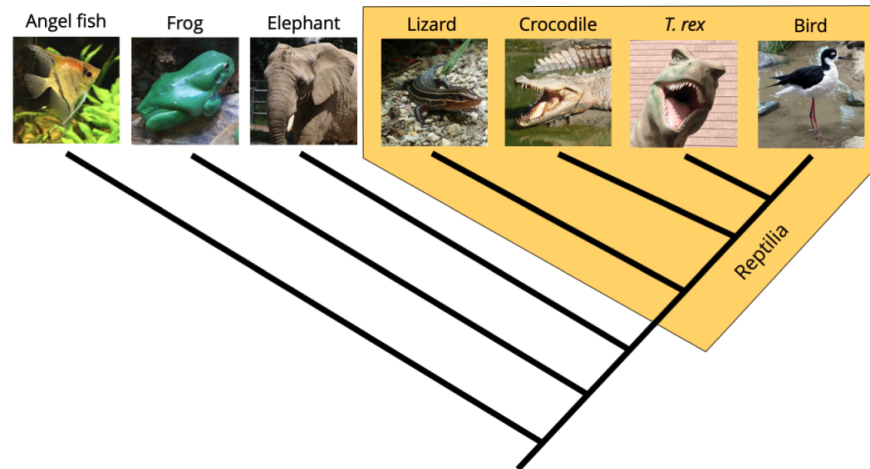


Figure 4: Evolutionary tree

- **Convergent evolution** refers to the independent evolution of similar phenotypic features in distinct lineages subjected to comparable selective pressures. Such convergently evolved traits are designated **analogous structures**, such as the wings of bats, birds, and butterflies.



Figure 5: Analogous structure

- (c) **The Fossil Record:** Paleontological evidence documents evolutionary change over geological timescales, providing a temporal dimension to our understanding of life's history.
- (d) **Biogeography:** The scientific study of species' geographic distributions reveals patterns consistent with evolutionary processes. Notably, oceanic islands typically harbor numerous endemic species—taxa found nowhere else on Earth—reflecting evolutionary diversification in geographic isolation.

2 Evolution of Populations

2.1 Genetic Variation and Equilibrium

Core Concept 2.1: Genetic Variation

Genetic variation encompasses the genetic differences among individuals within a **population**. This variation arises through multiple mechanisms, including the formation of new alleles via **mutation** (Core Concept 9.17), alterations in gene number or chromosomal position, and genetic recombination during **sexual reproduction** (Core Concept 1.2). While most mutations are deleterious or neutral, a small fraction confer selective advantages and serve as the raw material for adaptive evolution.



Figure 6: Phenotypic variation in horses

In evolutionary contexts, a **population** is defined as a group of individuals of the same species that occupy a common geographic area and interbreed, producing **fertile** offspring. Collectively, the **gene pool** represents the complete genetic composition of a population, encompassing all alleles present at all genetic loci.

Core Concept 2.2: Hardy-Weinberg Equilibrium

Hardy-Weinberg equilibrium describes a theoretical state in which a population is **not evolving**; allele and genotype frequencies remain constant across generations, provided that only Mendelian segregation and recombination operate. Under this model, if the frequency of the dominant allele is denoted as p and the frequency of the recessive allele as q , then

$$p + q = 1$$

representing all alleles in the population. Genotype frequencies are described by

$$p^2 + 2pq + q^2 = 1$$

where p^2 , $2pq$, and q^2 represent the frequencies of homozygous dominant, heterozygous, and homozygous recessive genotypes, respectively.

2.2 Mechanisms of Evolution

Core Concept 2.3: Adaptive Evolution

As briefly defined in Core Concept 1.2, natural selection operates by causing alleles to be transmitted to subsequent generations in proportions that differ from those observed in the parental generation.

Adaptive evolution is the process whereby traits that **enhance organismal survival or reproductive success** tend to increase in frequency within populations over successive generations.

Core Concept 2.4: Genetic Drift

Genetic drift is a random evolutionary mechanism whereby chance events cause allele frequencies to fluctuate unpredictably from one generation to the next, **independently of natural selection**. Genetic drift most commonly occurs via the following two events.

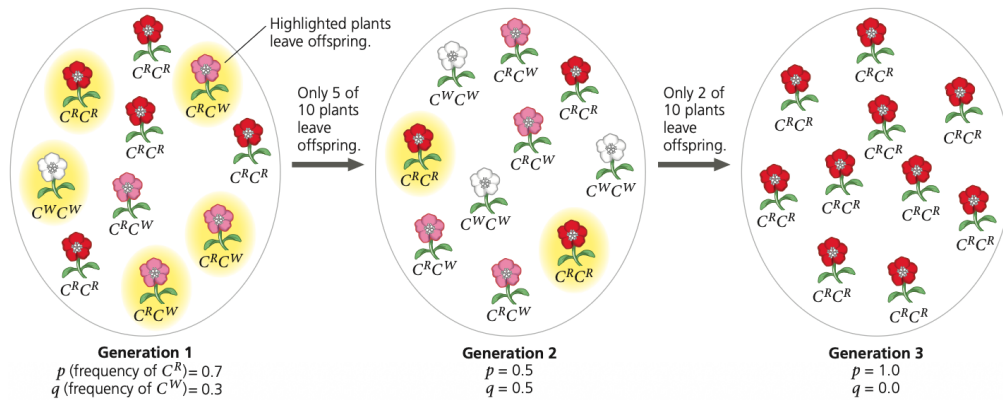


Figure 7: Genetic drift within a theoretical flower population

- (a) **Founder effect:** When a small number of individuals become isolated from a larger source population, the colonizing group may establish a new population whose gene pool differs substantially from that of the source population due to **sampling error**.

For example, when a few individuals are dispersed by storm events to colonize a remote island, the genetic composition of the founding population may not be representative of the parental population.

- (b) **Bottleneck effect:** A sudden environmental perturbation—such as fire, flood, or disease outbreak—may drastically reduce population size, resulting in a corresponding reduction in genetic diversity as only a subset of the original gene pool survives.

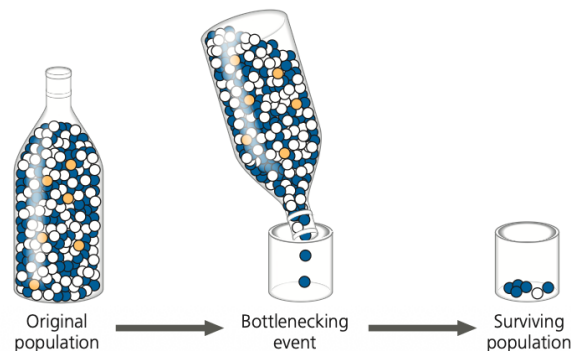


Figure 8: Bottleneck effect

The key distinction between the founder and bottleneck effect is that the former involves geographic dispersal and colonization, whereas the latter results from demographic catastrophe within an existing population.

The evolutionary consequences of genetic drift include the following.

- Genetic drift exerts stronger effects in small populations, where random sampling errors have proportionally greater impact on allele frequencies.
- Genetic drift causes allele frequencies to change randomly, *without* regard to the adaptive value of alleles.
- Genetic drift can result in less genetic variation within populations through the random loss of alleles.
- Genetic drift can result in the fixation of deleterious alleles, particularly in small populations where selection is less effective.

Core Concept 2.5: Gene Flow

Gene flow is the transfer of alleles into or out of a population resulting from the **movement of fertile individuals or their gametes**. Gene flow increases genetic variation within the recipient population by introducing new alleles.

*In contrast to the founder effect, which involves a one-time colonization event by a small, genetically impoverished subset of individuals resulting in **reduced** genetic diversity, gene flow represents continuous genetic exchange that actively **enhances** genetic variation in recipient populations.*

2.3 Fine-Tuning Natural Selection**Core Concept 2.6: Types of Natural Selection**

Natural selection is the only evolutionary mechanism that consistently produces adaptive evolution—that is, evolutionary change that increases organismal fitness in a given environment. The measure of this fitness differential is captured by **relative fitness**, which quantifies the contribution an individual makes to the gene pool of the subsequent generation relative to the contributions of other individuals within the population. Depending on the relationship between phenotype and fitness, natural selection operates through several distinct modes.

- (a) **Directional selection:** Environmental conditions favor individuals exhibiting one extreme of the phenotypic distribution, resulting in a shift of the population mean toward that extreme (*graphically represented as a shift of the frequency distribution curve in one direction*).

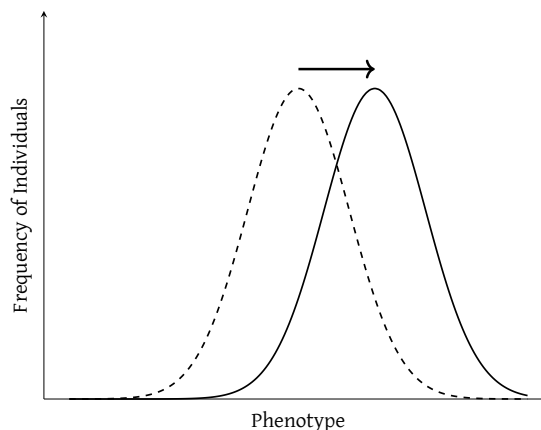


Figure 9: Directional selection

- (b) **Disruptive selection:** Environmental conditions favor individuals at both extremes of the phenotypic distribution while selecting against intermediate phenotypes, resulting in a bimodal frequency distribution.

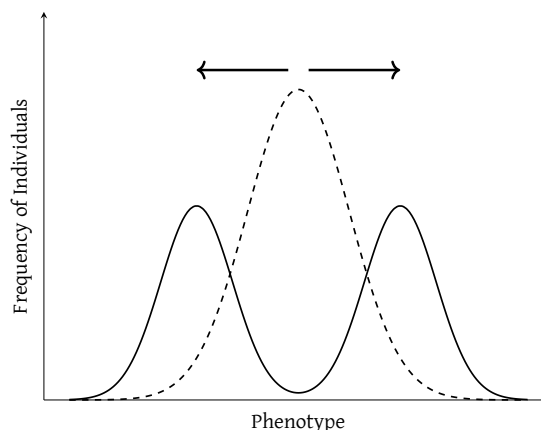


Figure 10: Disruptive selection

- (c) **Stabilizing selection:** Selection acts against both phenotypic extremes while favoring intermediate variants, thereby reducing phenotypic variance within the population (*graphically represented as a narrowing of the frequency distribution curve*).

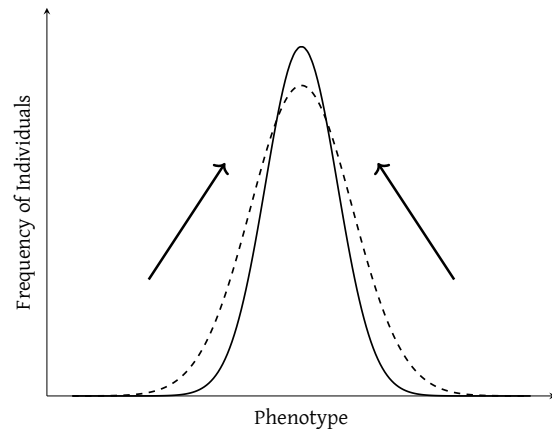


Figure 11: Stabilizing selection

Core Concept 2.7: Types of Sexual Selection

Sexual selection is an evolutionary process whereby individuals possessing certain inherited characteristics achieve greater mating success than other individuals of the same sex. This form of selection frequently produces **sexual dimorphism**, phenotypic differences in secondary sexual characteristics between males and females of the same species. Sexual selection can also be categorized into distinct modes.

- (a) **Intrasexual selection:** Individuals of one sex (*typically males*) compete directly with one another for access to mates of the opposite sex.

For example, male lions exclude rival males from mating opportunities with females in their pride.



Figure 12: Intrasexual selection

- (b) **Intersexual selection:** Individuals of one sex (*typically females*) exercise mate choice, selectively mating with individuals of the opposite sex based on specific trait preferences.

This mechanism often drives the evolution of elaborate male ornaments or displays, as exemplified by the ornate plumage patterns exhibited by male peacocks.



Figure 13: Intersexual selection

Core Concept 2.8: Balancing Selection

Balancing selection encompasses selective regimes that actively maintain genetic variation at particular loci within populations.

- (a) **Frequency-Dependent Selection:** The fitness of a phenotype is contingent upon its frequency within the population.

For example, left-mouthed and right-mouthed morphs of scale-eating cichlid fish maintain approximately equal frequencies because increased prevalence of either morph intensifies intramorph competition, thereby conferring a relative advantage to the rarer morph.

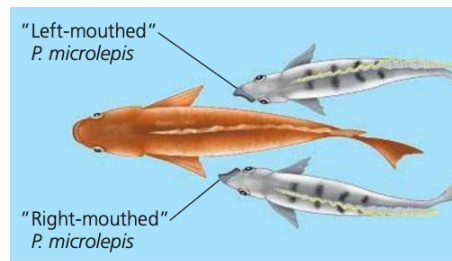


Figure 14: Frequency dependent selection

- (b) **Heterozygote Advantage:** Heterozygous individuals at a particular locus exhibit higher fitness than either homozygous genotype.

A classic example involves the human hemoglobin locus: homozygous dominant individuals face elevated risk of severe malarial infection and mortality, homozygous recessive individuals develop sickle cell anemia, whereas heterozygous individuals neither suffer from sickle cell anemia nor succumb readily to malaria, as the presence of some sickle-shaped erythrocytes impedes malarial parasite development.

Core Concept 2.9: Limitations of Natural Selection

Natural selection does not produce perfectly optimized organisms due to several fundamental constraints.

- Selection can act only upon **existing phenotypic variation**; it *cannot* generate novel variants de novo.
- Evolutionary trajectories are **constrained by phylogenetic history** and developmental architecture inherited from ancestral lineages.
- Adaptations frequently represent **evolutionary compromises** among conflicting selective pressures.

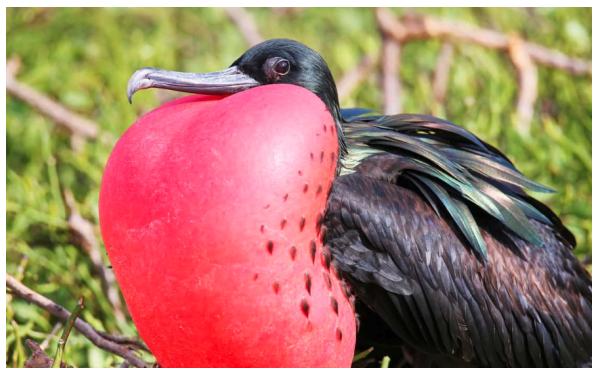


Figure 15: Evolutionary compromise of frigatebird gular sac between sexual selection and predatory pressures

For example, the inflatable red gular sac displayed by male frigatebirds (Figure 15) enhances mating success through sexual selection but simultaneously increases exposure to predators.

- Evolutionary outcomes reflect **complex interactions** among random events, natural selection, and environmental variation.

3 Species

3.1 Defining Species

Core Concept 3.1: Species & Speciation

Speciation is the evolutionary process by which a single ancestral species diverges into two or more descendant species.

Speciation constitutes a conceptual bridge linking **microevolution**—temporal changes in allele frequencies within populations—to **macroevolution**, the large-scale patterns of evolutionary diversification observed above the species level.

The **biological species concept** is the primary operational definition of a species among scientists. It states that a species comprises a group of populations whose members possess the **potential to interbreed** under natural conditions and produce **viable, fertile offspring**, but do not successfully interbreed with members of other such groups.

Core Concept 3.2: Other Definitions of Species

Alternative species concepts have been proposed to address limitations of the biological species concept.

- **Morphological Species Concept:** Species are distinguished on the basis of morphological characteristics such as body form and structural features. A limitation of this approach is its reliance upon potentially subjective morphological criteria.
- **Ecological Species Concept:** Species are defined by their ecological niche—the comprehensive set of interactions between members of the species and both the abiotic and biotic components of their environment.

Core Concept 3.3: Reproductive Barriers

Reproductive barriers are biological mechanisms that prevent gene flow between populations and thereby maintain species boundaries. These barriers are classified temporally relative to fertilization.

(a) Prezygotic Barriers (*pre-mating*)

- **Habitat isolation:** Populations occupy distinct habitats and therefore rarely encounter one another.
- **Temporal isolation:** Populations breed during different times (*daily, seasonally, or annually*), preventing reproductive contact.
- **Behavioral isolation:** Species-specific courtship rituals, mate attraction behaviors, and other behavioral repertoires prevent interspecific mating.

(b) Prezygotic Barriers (*post-mating, pre-fertilization*)

- **Mechanical isolation:** Morphological incompatibilities prevent successful copulation or gamete transfer.
- **Gametic isolation:** Sperm of one species fail to fertilize eggs of another species due to gamete incompatibility, biochemical barriers, or reduced sperm viability in foreign reproductive tracts.

(c) Postzygotic Barriers (*post-fertilization*)

- **Reduced hybrid viability:** Genetic incompatibilities between parental genomes impair hybrid development or survival.

- **Reduced hybrid fertility:** Hybrid offspring are partially or completely infertile.



Figure 16: Reduced hybrid fertility in mules

For example, the hybrid offspring produced by crossing a male donkey with a female horse yields a mule, an organism that exhibits robustness yet remains reproductively sterile.

- **Hybrid breakdown:** First-generation (F_1) hybrids exhibit normal viability and fertility, but subsequent generations (F_2 and beyond) display reduced fitness or sterility.

3.2 Mechanisms of Speciation

Core Concept 3.4: Allopatric Speciation

Allopatric speciation occurs when gene flow is interrupted by the geographic separation of a population into isolated subpopulations.

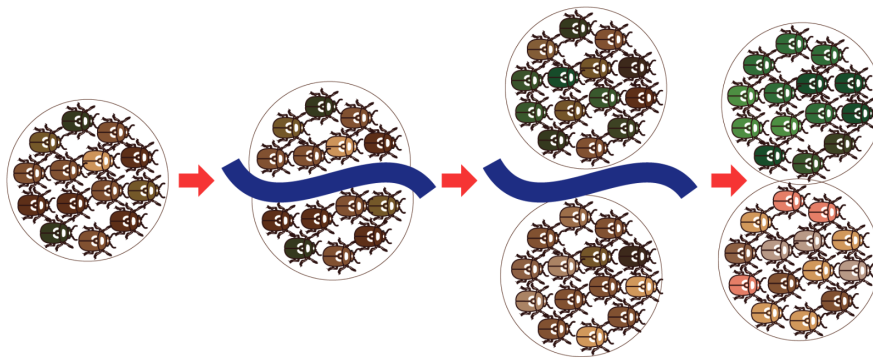


Figure 17: Allopatric speciation

For example, fish populations originally belonging to a single species may become geographically isolated in separate ponds differing in predator presence. The population subjected to predation evolves morphologies favoring rapid burst swimming, whereas the predator-free population evolves traits conducive to sustained swimming. Even when these populations are subsequently brought into geographic contact, evolved behavioral differences may constitute reproductive barriers, maintaining species boundaries.

Core Concept 3.5: Sympatric Speciation

Sympatric speciation is the evolution of reproductive isolation within populations occupying the same geographic area, occurring without geographic barriers to gene flow. Sympatric speciation may arise through several mechanisms.

- **Chromosomal aberrations** during cell division, particularly polyploidy (*Core Concept 3.8*), which creates immediate reproductive isolation.
- **Divergent sexual selection** may drive pronounced phenotypic divergence within initially mating populations.
- **Ecological specialization** involving shifts to novel habitats or resources not exploited by the parental population, potentially resulting in disruptive selection and eventual reproductive isolation.

3.3 Dynamics and Kinetics of Speciation

Core Concept 3.6: Hybrid Zones

Hybrid zones are geographic regions in which members of distinct species or divergent populations encounter one another and interbreed, producing at least some offspring of mixed ancestry. Hybrid zones may exhibit three distinct evolutionary trajectories over time.

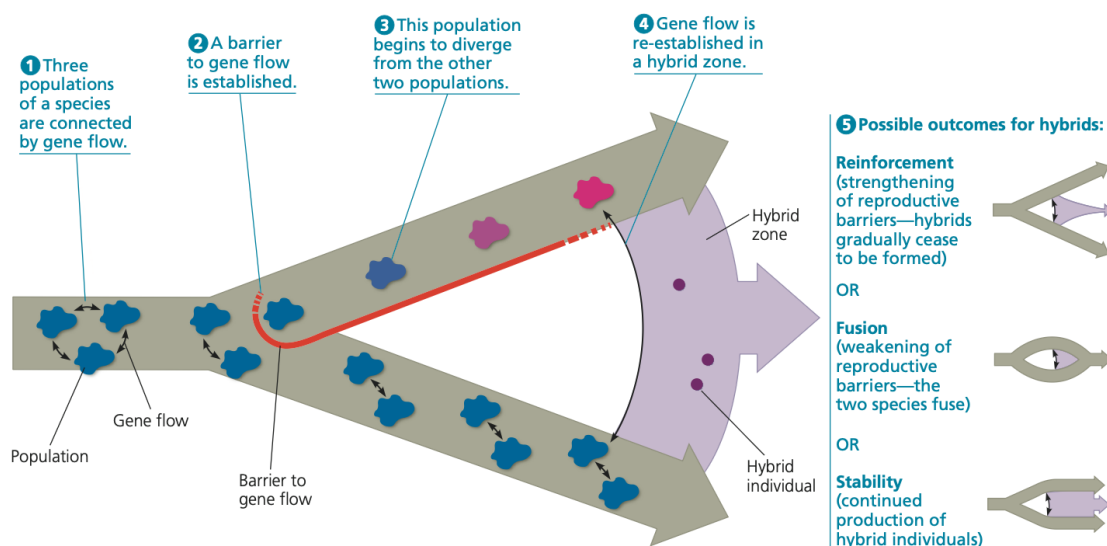


Figure 18: Evolutionary trajectories of hybrid zones

- Reinforcement:** When hybrid offspring exhibit reduced fitness relative to parental species, natural selection favors strengthening of prezygotic reproductive barriers, thereby reducing the frequency of maladaptive hybridization events. This process **reinforces species boundaries**.
- Fusion:** When reproductive barriers are weak or hybrid fitness is comparable to that of parental types, extensive gene flow may occur across the hybrid zone. Continued introgression may further erode reproductive barriers, causing the gene pools of the two species to converge. This reversal of the speciation process ultimately results in the **fusion of previously distinct species into a single interbreeding population**.

- (c) **Stability:** Hybrid zones may persist indefinitely with continued production of hybrid individuals, representing a stable equilibrium between gene flow and selection.

Core Concept 3.7: Speciation Models

Speciation may proceed at vastly different temporal rates, giving rise to two contrasting models.

- **Punctuated Equilibrium Model:** Most morphological and genetic divergence occurs during the speciation event itself, with newly formed species exhibiting relatively little evolutionary change throughout the remainder of their existence.

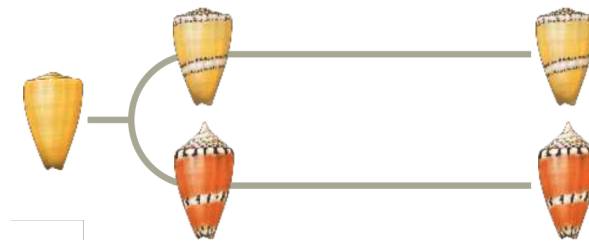


Figure 19: Punctuated equilibrium model

- **Gradualism Model:** Species diverge from one another through the slow, continuous accumulation of evolutionary changes over extended periods, producing a more uniform tempo of morphological change throughout a lineage's history.

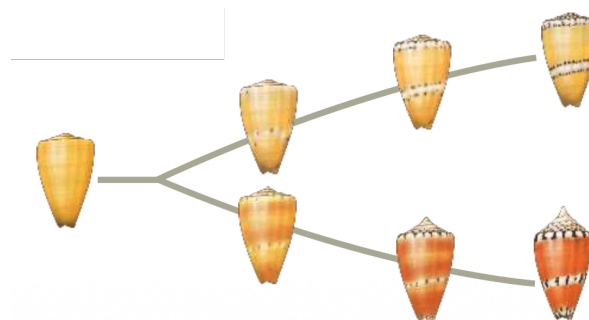


Figure 20: Gradualism model